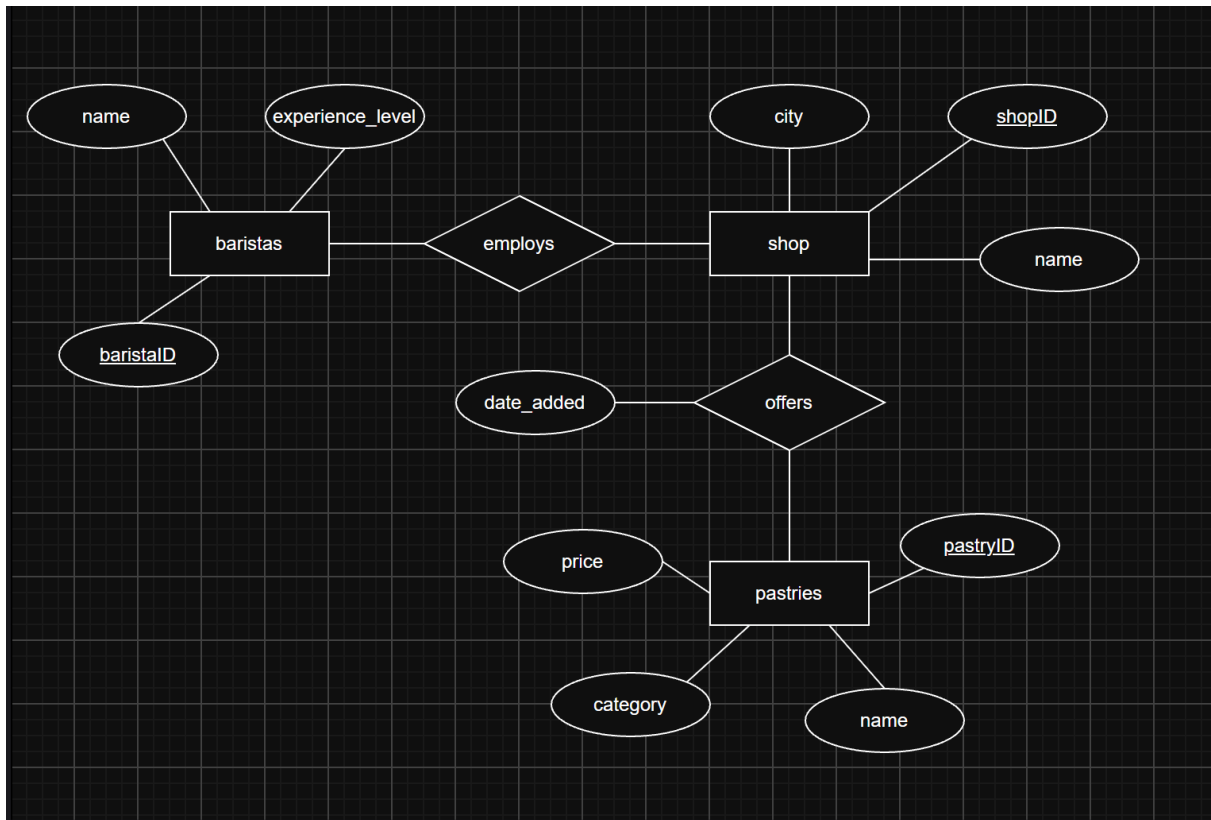


Krishna Santhosh
HW2



```
CREATE TABLE baristas (  
    baristaID INT PRIMARY KEY,  
    name VARCHAR(40),  
    experience_level VARCHAR(40)  
);
```

```
CREATE TABLE shops(  
    shopID INT PRIMARY KEY,  
    name VARCHAR(40),  
    city VARCHAR(40)  
);
```

```
CREATE TABLE employs(  
    baristaID INT ,  
    shopID INT,  
    FOREIGN KEY(baristaID) REFERENCES baristas(baristaID),  
    FOREIGN KEY(shopID) REFERENCES shops(shopID)  
);
```

```
CREATE TABLE pastries(  
    pastryID INT PRIMARY KEY,  
    name VARCHAR(40),  
    category VARCHAR(40),  
    price DOUBLE  
);
```

```
CREATE TABLE offers(  
    shopID INT,  
    pastryID INT,  
    date_added VARCHAR(40),  
    FOREIGN KEY(shopID) REFERENCES shops(shopID),  
    FOREIGN KEY(pastryID) REFERENCES pastries(pastryID)  
);
```

```
-- 1  
SELECT category, AVG(price)v  
FROM pastries  
GROUP BY category;
```

```

37      -- 1
38      • SELECT category, AVG(price)
39      FROM pastries
40      GROUP BY category;
41
42
43      -- 2

```

category	AVG(price)
Pastry	4.666666666666667
Muffin	3.625
Bakery	4.25
Scone	3.85
Breakfast	8
Cake	3.875

-- 2

```

SELECT experience_level, COUNT(*)
FROM baristas
GROUP BY experience_level;

```

```

43      -- 2
44      • SELECT experience_level, COUNT(*)
45      FROM baristas
46      GROUP BY experience_level;
47
48
49      -- 3.

```

experience_level	COUNT(*)
Junior	2
Senior	2
Mid-Level	2

-- 3.

```

SELECT city, COUNT(*)
FROM shops
GROUP BY city;

```

```

47
48
49      -- 3.
50      • SELECT city, COUNT(*)
51      FROM shops
52      GROUP BY city;
53

```

city	COUNT(*)
Seattle	1
Portland	1
Austin	1
Chicago	1
Denver	1
Boston	1

-- 4.

```
SELECT category, MAX(PRICE)
FROM pastries
GROUP BY category;
```

53
54
55 -- 4.
56 • SELECT category, MAX(PRICE)
57 FROM pastries
58 GROUP BY category;
59

category	MAX(PRICE)
Pastry	5.5
Muffin	3.75
Bakery	4.25
Scone	3.85
Breakfast	8.5
Cake	4

-- 5.

```
SELECT shopID, COUNT(*)
FROM offers
GROUP BY shopID;
```

59
60
61 -- 5.
62 • SELECT shopID, COUNT(*)
63 FROM offers
64 GROUP BY shopID;
65

shopID	COUNT(*)
1	2
2	2
3	2
4	2
5	2
6	2

-- 6.

```
SELECT p.name, p.category, p.price
FROM pastries p
JOIN (
    SELECT category, MAX(price) AS max_price
    FROM pastries
    GROUP BY category
```

) AS category_max

ON p.category = category_max.category AND p.price = category_max.max_price;

```
74  ) AS category_max
75  ON p.category = category_m
76
```

Result Grid	Filter Rows:																								
▶	<table> <thead> <tr> <th>name</th> <th>category</th> <th>price</th> </tr> </thead> <tbody> <tr> <td>Almond Croissant</td> <td>Pastry</td> <td>5.5</td> </tr> <tr> <td>Blueberry Muffin</td> <td>Muffin</td> <td>3.75</td> </tr> <tr> <td>Cinnamon Roll</td> <td>Bakery</td> <td>4.25</td> </tr> <tr> <td>Chocolate Scone</td> <td>Scone</td> <td>3.85</td> </tr> <tr> <td>Blueberry Scone</td> <td>Scone</td> <td>3.85</td> </tr> <tr> <td>Breakfast Burrito</td> <td>Breakfast</td> <td>8.5</td> </tr> <tr> <td>Lemon Loaf</td> <td>Cake</td> <td>4</td> </tr> </tbody> </table>	name	category	price	Almond Croissant	Pastry	5.5	Blueberry Muffin	Muffin	3.75	Cinnamon Roll	Bakery	4.25	Chocolate Scone	Scone	3.85	Blueberry Scone	Scone	3.85	Breakfast Burrito	Breakfast	8.5	Lemon Loaf	Cake	4
name	category	price																							
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Blueberry Scone	Scone	3.85																							
Breakfast Burrito	Breakfast	8.5																							
Lemon Loaf	Cake	4																							

-- 7.

SELECT DISTINCT offers.shopID

FROM offers

JOIN pastries p ON offers.pastryID = p.pastryID

WHERE p.price > (SELECT AVG(price) FROM pastries);

```
77
78  -- 7.
79  • SELECT DISTINCT offers.shopID
80  FROM offers
81  JOIN pastries p ON offers.pastryID = p.pastryID
82  WHERE p.price > (SELECT AVG(price) FROM pastries);
83
```

shopID
1
5

-- 8.

SELECT shopID, pastryID

FROM offers

WHERE date_added = (SELECT MIN(date_added) FROM offers);

```
83
84
85  -- 8.
86  • SELECT shopID, pastryID
87  FROM offers
88  WHERE date_added = (SELECT MIN(date_added) FROM offers);
89
```

shopID	pastryID
1	1

-- 9.

```
SELECT shopID, COUNT(*) AS total_pastries
FROM offers
GROUP BY shopID
HAVING COUNT(*) = (
    SELECT MAX(shop_counts.cnt)
    FROM (
        SELECT COUNT(*) AS cnt
        FROM offers
        GROUP BY shopID
    ) AS shop_counts
);
```

	shopID	total_pastries
▶	1	2
	2	2
	3	2
	4	2
	5	2
	6	2

-- 10.

```
SELECT name FROM baristas
WHERE baristaID IN (
    SELECT baristaID
    FROM employs
    WHERE shopID IN (
        SELECT shopID
        FROM shops
        WHERE city = 'Seattle'
    )
);
```

116

117

Result Grid	
	name
▶	Emma Watson
	Liam Chen